Mustafa Rezk *Data Engineer*

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PROFILE

Data Engineer experienced in building scalable ETL/ELT pipelines using Python, SQL, Apache Spark, and Airflow, with hands-on experience in Microsoft Fabric and SAP data integration.Skilled in Medallion Architecture, data modeling, data warehousing, lakehouse concepts, and query optimization for enterprise analytics and BI.

EDUCATION

Cairo University, *Master Data science* 09/2026 – Present

Higher Institute for Engineering and Technology (HIET), Egypt, 09/2019 – 08/2024
Bachelor's in Computer and Control Engineering

SKILLS

Languages & Databases: Python,SQL, Bash | **Data Engineering:** ETL/ELT, Airflow, dbt, Kafka, Data Modeling, Pyspark, Databricks | **BI & Visualization Tools:** Power BI, Locker Studio , Tableau | **DevOps:** Docker, Git/CI-CD | **Cloud Platfrom:** Microsoft Fabric (Lakehouse, Dataflows, Pipelines), AWS (S3, Redshift, Glue, Athena, Lambda), GCP, Snowflake

PROFESSIONAL EXPERIENCE

Ai & Data Management, *DTC* 09/2025 – Present
- Contributed to the AI team's R&D efforts for an AI-powered SaaS platform, helping develop a Retrieval-Augmented Generation (RAG) system.
- Collaborated with team members to translate requirements into platform features and support development priorities.
- Applied data governance and DAMA principles to improve data definitions, ownership, standardization, and quality.
- Collaborated with technical and business stakeholders to understand data requirements and support reliable analytical datasets.

Data Engineering Instructor & Graduation Project Mentor, *Information Technology Institute (ITI)* 07/2026 – Present
- Delivered hands-on training in Data Warehousing (DWH), dimensional modeling, ETL/ELT design, and distributed data processing using Apache Spark.
- Mentored an ITI 9-Month Data Management Professional Training Program graduation project team, providing technical guidance, project architecture support.

INTERNSHIPS

Information Technology Institute (ITI), *Data Engineering Track* 03/2025 – 07/2025
- Completed intensive training across the modern data stack: DWH, ETL/ELT, Informatica, Hadoop ecosystem (HDFS, Hive), Apache Spark, and Databricks
- Applied dimensional modeling and big data processing concepts through hands-on labs and mini-projects covering batch and distributed pipelines
- Built foundational proficiency in data warehouse architecture, data integration, and cloud-based data engineering tooling

Digital Egypt Pioneers Initiative (DEPI), 07/2024 – 12/2024
Data Science Track (Egyptian Ministry of Communications and Information Technology)
- Mastered the data science lifecycle including rigorous Exploratory Data Analysis (EDA), feature engineering, and advanced data cleansing techniques.
- Developed and optimized Machine Learning and Deep Learning algorithms for complex classification and regression problems.

PROJECTS

End-to-End Data Platform on Microsoft Fabric (Medallion Architecture),
Microsoft Fabric · Data Pipelines · Lakehouse · PySpark · Fabric Warehouse · Power BI
- Built an automated Fabric Data Factory pipeline (HTTP, Lookup, ForEach, Copy) that ingests 8 datasets into a Lakehouse Bronze layer, scheduled daily with Outlook/Teams alerts.
- Cleaned and validated data in the Silver layer with PySpark and Delta Lake (deduplication, null handling, key checks), then loaded it incrementally into a Warehouse using a watermark (CDC).
- Modeled a star schema in the Gold layer and connected Power BI to build a sales dashboard covering 18.5M in sales and 7.9M in profit.

Real-time Agricultural IoT Data Pipeline & Analytics Solution(AWS)
- Engineered a real-time data pipeline using AWS IoT Core and Lambda to ingest and process sensor data and Weather APIs.
- Integrated Amazon Redshift as a centralized Data Warehouse for historical analysis, implementing efficient schema designs for time-series data.
- Established full-stack observability using CloudWatch and OpenSearch, optimizing irrigation cycles based on real-time metrics and predictive data.

Sales Data Warehouse & Analytics

- Designed a retail data warehouse using conceptual, logical, and physical data modeling principles.
- Created a dimensional model with fact and dimension tables using star-schema design for sales reporting.
- Integrated more than 100,000 CRM and ERP records into a centralized SQL Server warehouse.
- Developed SQL queries for data profiling, validation, reconciliation, duplicate detection, null-value checks, and referential-integrity testing.
- Built ETL transformations for cleansing, standardization, deduplication, and business-rule validation.
- Improved data quality to 99% by resolving duplicate, missing, and inconsistent records.

CERTIFICATES

- AWS Certified Cloud Practitioner
- IBM Data Engineering Professional Certificate - 2025
- Salesforce Certified Tableau Desktop Foundations